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# LexSR: A Legal Knowledge Graph for a Low-Resource Jurisdiction — Consolidating Legislation and Linking Case Law for Suriname

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**Abstract.** Access to consolidated, machine-readable legislation and linked case law is a prerequisite for effective legal information systems. For Suriname — a Dutch-speaking jurisdiction of fewer than 600,000 people with a predominantly scanned legal archive — no such resource previously existed. We present LexSR, a pipeline that ingests raw government PDFs and the Hof van Justitie's case-law portal, applies OCR, classifies documents, automatically consolidates *wijzigingswetten* (amendment acts) into versioned legislative expressions modelled on FRBR, and links case law to the legislation it applies. The pipeline produces a corpus of 1,623 works (1,323 legislative, 300 judicial) and 45,293 fragments, from which we construct a unified knowledge graph of 2,768 edges spanning amendment, citation, and case-application relationships. We report two evaluations: a synthetic citation-resolution benchmark, where our resolver achieves 100% precision/recall against a curated 61-case gold set, and — the paper's methodological contribution — a measurement of how much of that resolution accuracy actually survives contact with the real, messily-titled corpus it was built to run against. Only 17% of extracted citations resolve to an in-corpus work; we trace this gap to a structural mismatch between idealized canonical identifiers and the filename-derived IDs that real-world scraped/OCR'd ingestion produces, and report the fix. We also measure amendment-parsing coverage as the corpus grew from 33 to 50 amendment documents (82% $\rightarrow$ 38%), a second honest signal of how benchmark accuracy on a curated subset can overstate real-corpus performance. We discuss implications for low-resource legal NLP and make the corpus, code, and evaluation scripts publicly available.

**Keywords.** legal knowledge graph, legislative consolidation, case-law citation linking, low-resource legal NLP, FRBR, Suriname

## 1. Introduction

Consolidation — merging an amendment act's instructions into the base law it modifies — and citation linkage — connecting case law to the provisions it applies — are both legally critical and technically demanding. Most European jurisdictions maintain authoritative consolidated legislation and case-law databases through governmental por-

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tals (e.g., *wetten.overheid.nl* and *rechtspraak.nl* in the Netherlands, EUR-Lex for EU law). Small and developing jurisdictions often lack the institutional capacity for either, leaving practitioners to manually track amendment chains and cross-reference case law with no integrated view of the law in force.

Suriname illustrates this gap sharply. The *Staatsblad* (official gazette) is available as individual scanned PDFs through several government sources, and the Hof van Justitie publishes case decisions on a dedicated portal (*rechtspraak.sr*, launched 2019), but no system links the two: a decision citing "artikel 4 Arbeidswet 1963" offers no way to jump to that provision, and no portal shows any act's consolidated text as amended.

This paper makes four contributions:

1. **LexSR corpus**: a structured, FRBR-modelled database of 1,323 Surinamese legislative works and 300 Hof van Justitie decisions, with 45,293 fragments, derived from ten live government sources, the judicial case-law portal, and locally digitised archives.
2. **Amendment parser**: a hybrid regex + LLM system that extracts structured consolidation instructions from *wijzigingswetten*, evaluated across the corpus's full growth trajectory.
3. **Case-law integration**: a scraper and ingestion pipeline for *rechtspraak.sr* that segments decisions into numbered considerations and links them to cited legislation, reusing the same citation-resolution machinery built for legislation-to-legislation citations.
4. **Knowledge graph and an honest accuracy audit**: a unified graph of amendment, citation, and case-application edges over the full corpus, together with a measurement of the gap between synthetic-benchmark citation accuracy and real-corpus linkage rate — and the fix that closed most of it.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes the corpus and ingestion pipeline, including the newly added case-law source. Section 4 details amendment consolidation. Section 5 describes case-law integration and knowledge graph construction. Section 6 presents evaluation results, including the citation-linkage accuracy audit. Section 7 discusses limitations and future work, and Section 8 concludes.

## 2. Related Work

### *2.1. Legal Document Modelling*

The FRBR (Functional Requirements for Bibliographic Records) model provides a well-established abstraction for legislative documents, distinguishing the abstract Work (a law), its Expression (a version at a given date), its Manifestation (a particular encoding), and individual Items [1]. Akoma Ntoso [2] is the primary XML schema for legislation structured around these concepts and has been adopted by the UN, EU, and several national parliaments. The European Legislation Identifier (ELI) [3] provides URI conventions compatible with this model. We extend this modelling to case law following the same FRBR pattern: a court decision is a Work with a single Expression (the published text does not get amended the way legislation does), and its numbered considerations are Fragments — the case-law analogue of legislative articles.

### 2.2. Legal Knowledge Systems for Consolidation

Automated consolidation has received sustained attention in the legal knowledge systems community. Biagioli, Francesconi et al. [4] describe the *Norme in Rete* project's XML standards and tools for cross-referencing and consolidating Italian legislation. Spinosa et al. [5] apply NLP-based metadata extraction to detect and classify textual amendments (repeal, substitution, integration) for the same consolidation task. The *legislation.gov.uk* system [6] handles consolidation for UK law at scale with a semi-automated editorial workflow. Vitali and Zeni [7] discuss amendment formalisation in Akoma Ntoso. The Dutch *wetten.overheid.nl* portal maintains consolidated texts for all Dutch and former-colonial legislation through a government editorial process, but does not publish its consolidation tooling [8]. LexSR differs from these in targeting a low-resource setting — no existing structured archive, predominantly scanned PDFs, and no editorial support — with a fully automated pipeline.

### 2.3. Legal Knowledge Graphs and Citation Networks

Citation network analysis of case law has a long history in AI and Law, from early work on precedent networks to more recent graph-based approaches to legal information retrieval and argumentation mining [9]. Legal knowledge graphs unifying legislation and case law have been built for well-resourced jurisdictions with existing structured data — e.g., Dhani et al. [10] construct one from Indian court judgments and statutes for case-similarity recommendation — but these rely on structured source data; we are not aware of prior work constructing such a graph from scratch for a jurisdiction whose primary source material is unstructured scanned PDFs and an unindexed WordPress-based case portal, which is the setting this paper addresses.

### 2.4. NLP for Legal Amendment Text

Information extraction from legal amendment provisions has been studied for several European languages. Lesmo, Mazzei, Palmirani et al. [11] present TULSI, an NLP system combining deep syntactic analysis with a shallow semantic interpreter to extract and classify modificatory provisions in Italian legislative texts. For Dutch, Boer et al. [12] propose the MetaLex/LKIF XML standards for structuring legislation, and Hoekstra et al. [13] extend this with the LKIF Core ontology of basic legal concepts, but explicit amendment/consolidation parsing for Dutch has received limited attention in the JURIX and ICAIL literature. The use of large language models for legal text understanding is well-established [14,15]. Our hybrid approach — regex-first with LLM fallback for low-confidence cases — is motivated by the need for determinism and auditability while retaining flexibility for novel constructions.

### 2.5. Legal Corpora for Small Jurisdictions

The JURIX and ICAIL communities have increasingly highlighted the need for legal NLP resources beyond the major European languages [16]. Krasadakis et al. [17] survey the specific challenges legal informatics poses for low-resource-language NLP. To our knowledge, no machine-readable structured corpus linking legislation and case law for Suriname has previously been published.

## 3. Corpus and Ingestion Pipeline

### *3.1. Sources*

LexSR ingests from ten live government sources plus locally digitised PDF archives: the parliament website (DNA.sr), the legal database SRIS, the tax authority (Belastingdienst), the central bank (MAS), the standards bureau (SSB), the main government portal, and four ministry sites (finance, labor, natural resources, and the electoral bureau). We evaluated all sixteen government sources our scraper infrastructure supports; six were excluded after diagnosis showed their configured URLs no longer match the live site structure (Section 7 discusses this as a recurring low-resource-jurisdiction problem: source sites change without notice and without a stable API to detect it). Case law is sourced separately from *rechtspraak.sr*, the Hof van Justitie's decision portal, described in Section 5.1. All source documents are stored immutably in a bronze layer with SHA-256 checksums and retrieval timestamps.

### *3.2. Data Model*

The canonical store implements FRBR at four levels, shown in Table 1.

**Table 1.** FRBR levels and LexSR entities

| FRBR level | LexSR entity | Legislation example | Case-law example |
|---|---|---|---|
| Work | `Work` | Arbeidswet 1963 | SRU-HvJ-2022-39 |
| Expression | `Expression` | version_date=2024-01-15 | single expression |
| Manifestation | `Manifestation` | source PDF | source HTML page |
| Item | `Fragment` | Artikel 4, lid 2 | consideration 2.1 |

Stable URIs follow an ELI-inspired pattern rooted at `/id/{scheme}/{id}`, with `law` and `caselaw` as the two schemes currently populated. The amendment relation is represented by an `AmendmentLink` table connecting an amending Work to the base Work it modifies; the citation relation (legislation-to-legislation, case-to-legislation, and case-to-case) is represented by a single generic `Citation` table typed by `citation_type`, avoiding a schema split between the two document kinds.

### *3.3. Corpus Statistics*

Table 2 summarizes the corpus. The legislative corpus spans material from 1931 to 2025, with the densest coverage in the 2010–2025 period corresponding to the availability of digital-native PDFs. The case-law sample spans 1952–2025 and was drawn from a discoverable pool of 900+ decisions on *rechtspraak.sr* (Section 5.1); this paper reports on the first 300 ingested.

### *3.4. OCR and Structure Extraction*

For scanned PDFs, text is extracted using a pipeline that applies standard PDF text extraction first and falls back to Tesseract OCR for image-based pages. Article boundaries are detected heuristically using the standard Dutch/Surinamese legislative structure: "Ar-

**Table 2.** Corpus statistics

| Metric | Value |
|---|---|
| Legislative works | 1,323 |
| Case-law works (decisions) | 300 |
| Expressions | 1,633 |
| Fragments (articles, paragraphs, considerations) | 45,293 |
| Considerations extracted from case law | 1,142 |
| Citations extracted | 15,867 |
| Citations resolved to an in-corpus work | 2,713 (17.1%) |
| Amendment links identified | 55 |
| Amendment links fully applied by the engine | 11 |
| Knowledge graph nodes / edges | 1,623 / 2,768 |

tikel N" headings, "Lid N" sub-paragraphs, and lettered or numbered sub-items. OCR quality varies substantially across the corpus; a practical, low-resource-specific finding is that the target machine did not have the Dutch Tesseract language pack installed, silently degrading a subset of the corpus to English-model OCR before this was caught and an availability check added (Section 7.1).

## 4. Amendment Consolidation

### *4.1. Structure of a* Wijzigingswet

A Surinamese amendment act (*wijzigingswet* or *wijzigingsbesluit*) follows a standard template. The operative section (Artikel 1 of the amendment) names the base law and lists labelled instructions:

```
Artikel 1
De Wet Toezicht en Controle op
Financiële Dienstverlening (S.B. 2009 no. 78)
wordt als volgt gewijzigd:

A. In artikel 3 lid 2 wordt "vijftien" vervangen door "tien".
B. Na artikel 16 wordt een nieuw artikel 16a
ingevoegd, luidende:
"Artikel 16a. ..."
C. Artikel 24 vervalt.
```

The amendment act's later articles (Artikel 2, 3, ...) may amend a second base law or specify the in-force date. The act typically concludes with a *Memorie van Toelichting* (explanatory memorandum), which must be excluded from instruction extraction.

### *4.2. Parser Architecture*

The `AmendmentParser` operates in two passes. **Pass 1 (regex)** splits the document into sections by "Artikel N" boundaries (after stripping explanatory-memorandum text), extracts the target law name from the section header or preamble, splits labelled instruction blocks (A., B., C. or Roman-numeral labels), and classifies each block against a priority-ordered set of $\sim$20 regex patterns covering the standard Dutch amendment constructions

(repeal, replacement, insertion, phrase substitution). **Pass 2 (LLM fallback)**, used only when a regex-classified instruction's confidence falls below 0.70, re-sends the section to an LLM with a structured JSON prompt; results are merged, with LLM output overriding regex output where its confidence is higher. This pass is disabled for the benchmark in Section 6.3 to report a clean regex-only baseline, consistent with our original evaluation methodology.

The LLM pass now supports both a local model (Qwen2.5-Coder-7B-Instruct via vLLM) and any OpenAI-compatible hosted chat endpoint, selected by URL shape. In practice we found hosted reasoning models require substantially larger token budgets than a plain completion model of similar size — a fixed 512–2048 token cap silently truncated every response, since part of the budget is consumed by hidden reasoning tokens before any answer content is emitted. We report this as a practical deployment note for others integrating hosted LLMs into constrained-latency legal NLP pipelines.

### *4.3. Consolidation Engine*

`ConsolidationEngine` consumes `AmendmentInstruction` objects and applies them to the current expression of the base law to produce a new `Expression` at the amendment's effective date. Base-law resolution — mapping the amendment's raw target-law reference to a Work in the store — uses fuzzy title matching (exact match, normalized match, then length-guarded substring containment) rather than exact-ID lookup, because the amendment's stated law name essentially never matches the filename-derived work ID that real ingestion produces. We factored this matching logic into a shared module once we discovered the citation resolver needed the identical fallback for the identical reason (Section 6.2).

## 5. Case-Law Integration and Knowledge Graph

### *5.1. Scraping* rechtspraak.sr

*rechtspraak.sr* is a WordPress-based site with no listing or search API; the only reliable enumeration mechanism is the WordPress XML sitemap, which we parse recursively (sitemap index → per-type sub-sitemaps → individual decision URLs). This alone surfaced 900+ decisions spanning 1952–2025. Each decision page exposes clean structured metadata (court, case number, decision and publication dates, legal area) as label/value pairs, and full decision text as HTML — considerably cleaner input than the OCR'd legislative PDFs, since no scanning or recognition step is involved.

### *5.2. Consideration Segmentation*

Surinamese court decisions structure their legal reasoning as numbered paragraphs under a "De beoordeling" (assessment) heading, using section.paragraph numbering (e.g., "2.1", "4.7") rather than an explicit "r.o." (*rechtsoverweging*) label. We segment on this numbering pattern directly; decisions without recognizable numbering (typically older or very short judgments) are kept as a single fragment rather than discarded, so no decision's text is lost to a structural-parsing failure. Across the 300 ingested decisions this produced 1,142 consideration-level fragments.

### *5.3. Citation Extraction from Case Law*

Legislative citations within case-law text ("artikel 4 lid 2 Arbeidswet 1963") are extracted using the identical citation resolver and extraction pipeline built for legislation-to-legislation citations — no case-law-specific resolution logic was needed, since a citation is a citation regardless of source document type. This yielded 526 case-to-legislation citations across the 300-decision sample. Case-to-case citations (a decision citing another by its site-assigned identifier, e.g. "SRU-HvJ-2019-3") use a separate lightweight regex match, since these identifiers are specific to this one source and not part of the general citation-resolution grammar; none were found in this sample, which we discuss as an open question in Section 7.1 rather than assume is a parsing gap.

### *5.4. Knowledge Graph Construction*

We assemble a unified graph directly from the canonical store — no additional schema or storage layer — with one node per Work (typed `legislation` or `case_law`) and three edge types: AMENDS (from `AmendmentLink`), CITES (legislation-to-legislation citations), and APPLIES (case-to-legislation citations). The resulting graph over the full corpus has 1,623 nodes and 2,768 edges (2,322 CITES, 391 APPLIES, 55 AMENDS), with a largest connected component of 706 nodes. The ten most-cited works are, in order: Personeelswet (civil service law), Burgerlijk Wetboek (Civil Code), Wetboek van Strafrecht (Criminal Code), Grondwet (Constitution), Wetboek van Koophandel (Commercial Code), and several tax and electoral statutes — a distribution that matches expectation for a general legal system and offers a face-validity check on the extraction pipeline independent of the benchmark in Section 6.1.

## 6. Evaluation

### *6.1. Citation Resolution: Synthetic Benchmark*

We evaluate the citation resolver on a curated 61-case known-answer dataset spanning citation, temporal, and gazette-reference categories. The resolver achieves a 100% pass rate, 100% precision/recall/F1, and 91.2% raw resolution rate (the remainder being deliberately-unresolvable negative test cases). This is unchanged from prior evaluation and confirms the resolver's pattern coverage is sound against well-formed input.

### *6.2. Citation Resolution: Real-Corpus Linkage Gap*

Synthetic-benchmark accuracy measures whether the resolver correctly identifies *that* a reference exists and what canonical URI it should map to. It does not measure whether that URI actually corresponds to anything in a real corpus — and for LexSR, at the time we built the knowledge graph, it did not: every one of 13,949 citations extracted from the corpus had a null target work reference, despite each having passed resolution. Three compounding causes, uncovered by inspecting the (previously unexamined) `target_work_id` column directly rather than the aggregate "citations resolved" count reported by the ingestion CLI:

1. **Stale identifier scheme.** 788 works and their expressions, ingested before a URI prefix convention was corrected mid-project, retained the old `sr/law/...` form instead of `/id/law/...`; direct URI lookup naturally failed for all of them. A one-time migration fixed this class.
2. **Idealized vs. filename-derived identifiers.** The resolver's abbreviation registry maps "Grondwet" to the canonical slug `grondwet-1987` — a clean identifier reflecting how the corpus should be organized. Real ingestion, run against scraped/OCR'd source documents with no canonical-naming authority, instead produces IDs derived from the source filename (`grondwet-met-mvt-4fb9cced`). No amount of resolver accuracy against a synthetic benchmark closes this gap, because the gap is not in resolution logic — it is a mismatch between two different, independently-reasonable identifier schemes. We closed most of it by reusing the fuzzy title-matching fallback already built for amendment base-law resolution (Section 4) as a fallback when direct URI lookup fails.
3. **Case-law scheme not recognized.** A citation-type classifier that checks source/target URIs for an `/ecli/` substring to detect case-law provenance silently mis-typed every citation originating from a case-law fragment as `unknown`, since this corpus's actual case-law scheme is `/id/caselaw/...` (Section 3), a straightforward but easy-to-miss consequence of extending a system originally built for one document type to a second.

After fixing all three, target-work resolution across the corpus rose from 0/13,949 to 2,713/15,867 (17.1%; the citation count itself grew between measurements as case-law ingestion added citations from decision text). The remaining 83% are overwhelmingly gazette references (*Staatsblad*/*Gouvernementsblad* citations), which are correctly extracted and typed but cannot resolve further because the corpus does not model individual gazette issues as first-class entities — a genuine scope boundary, not a resolution failure, and one we flag as future work (Section 7.3) rather than paper over.

We report this gap because we believe it generalizes: a resolver validated only against a synthetic or curated benchmark can report perfect accuracy while linking nothing in the corpus it is meant to serve, if the benchmark's identifiers do not reflect how real ingestion actually names things. We are not aware of this specific failure mode being discussed in the low-resource legal NLP literature and suggest it as a checklist item for similar projects: *measure resolution against the real corpus's actual identifiers, not just against curated ground truth.*

### *6.3. Amendment Parsing: Coverage Over Corpus Growth*

We evaluate the amendment parser (regex pass only, matching the methodology of our earlier 33-document evaluation) on all amendment documents currently in the corpus (a document may target more than one base law, so this is evaluated per amendment document, not per `AmendmentLink` row — the corpus has 55 links across 50 unique documents). Table 3 reports both measurements.

The drop is a second instance of the same underlying lesson as Section 6.2: the original 33-document set was disproportionately drawn from cleanly-digitised, well-OCR'd sources, while the expanded corpus includes material from ten additional live-scraped sources with more variable OCR quality and, in some cases, genuinely different amendment-drafting conventions we have not yet characterized. As with the original

**Table 3.** Amendment parsing coverage before and after corpus expansion

| Corpus state | Amendment documents | OK (≥1 instruction) | Rate |
|---|---|---|---|
| Original (curated subset) | 33 | 27 | 81.8% |
| Current (full live-scraped corpus) | 50 | 19 | 38.0% |

evaluation, failures are concentrated in cases where the operative instruction text itself is missing or garbled in the source fragment, not in cases where legible operative text was misparsed — but we have not yet re-run the original's per-case failure classification (OCR garbling vs. missing text vs. novel construction) at the new scale, and flag this as immediate future work (Section 7.3) rather than assert a cause we have not verified at this corpus size.

### *6.4. Engine Validation*

Eleven amendments have been fully applied by the consolidation engine (up from 2 at the prior evaluation) and are available for inspection in the canonical store; the remainder of the 19 parseable-but-unapplied cases fail engine-level base-law resolution or produce low-confidence instructions withheld pending review, rather than being blocked by parsing.

## 7. Discussion and Future Work

### *7.1. Limitations*

**Corpus coverage and source stability.** Six of sixteen candidate scraper sources were excluded because their configured URLs no longer resolve or the site structure has changed — a specific instance of a general low-resource-jurisdiction problem: government sites are maintained inconsistently and rarely expose a stable API, so scraper configurations decay silently over time with no error signal beyond "zero documents found." Automated periodic health-checking of scraper coverage is future infrastructure work, not yet built.

**Real-corpus linkage.** Section 6.2's 17.1% real-corpus citation linkage rate, while a large improvement over the pre-fix 0%, remains low in absolute terms; the majority of the gap is the un-modelled gazette-issue scope boundary rather than a resolution failure, but this should be stated as a limitation of the current corpus's entity coverage, not treated as solved.

**Amendment-parsing coverage.** The 38% current-corpus OK rate (Section 6.3) is a lower bar than we would want for a production consolidation system, though it should be read alongside the fact that "OK" here means at least one clean instruction was extracted per amendment — many practically important amendments are simpler, single-instruction acts well within this capability.

**Case-to-case citation coverage.** Zero case-to-case citations were found in the 300-decision sample (Section 5); we do not yet know whether this reflects a genuine rarity of explicit cross-citation in this corpus (plausible for a small jurisdiction with a correspondingly small body of precedent) or an extraction-pattern gap, and consider this an open methodological question rather than a settled negative result.

### *7.2. Broader Implications*

Suriname shares its legislative drafting conventions (Dutch-language templates derived from former colonial law) with the Dutch Caribbean islands (Aruba, Curaçao, Sint Maarten). The LexSR pipeline may be directly applicable to these jurisdictions with minimal adaptation. More broadly, the paper's central methodological point — that synthetic benchmark accuracy and real-corpus linkage rate are different measurements that can diverge sharply — applies to any legal knowledge system built by combining a hand-curated resolution component with uncurated, real-world ingestion, independent of jurisdiction or language.

### *7.3. Future Work*

1. **Failure-mode analysis at scale**: Re-run the original's per-case classification (OCR garbling vs. missing operative text vs. novel construction) across the current 50-document amendment corpus.
2. **Gazette-issue modelling**: Represent individual *Staatsblad*/*Gouvernementsblad* issues as first-class entities to close the largest remaining share of the citation-linkage gap.
3. **Case-law scale-up**: Ingest the remaining 600+ decisions available on *rechtspraak.sr* and re-evaluate case-to-case citation coverage at that scale.
4. **Gold-standard instruction annotation**: Annotate amendment acts at instruction level (target article, operation type, find/replace text) for precision/recall evaluation beyond the current binary OK/NO_MATCH measure.
5. **Source health monitoring**: Automated detection of scraper configuration decay (Section 7.1) rather than manual discovery.
6. **User evaluation**: Study practitioner use of the portal and knowledge graph through a user study with Surinamese legal practitioners.

## 8. Conclusion

We presented LexSR, a pipeline for building a structured, versioned corpus of Surinamese legislation and case law, automatically consolidating amendment acts, and constructing a unified knowledge graph linking legislation, amendments, and case-law citations. Beyond the system itself, this paper's central contribution is methodological: we measured, rather than assumed, how much of our citation resolver's synthetic-benchmark accuracy (100% precision/ recall) survives contact with a real, uncurated corpus (17.1% linkage), traced the gap to a structural mismatch between idealized and filename-derived identifiers, and report the fix. A parallel measurement of amendment-parsing coverage across corpus growth (82% $\rightarrow$ 38%) tells the same story from a different angle. We believe both findings generalize beyond this jurisdiction: legal NLP systems evaluated only against curated benchmarks can silently fail to serve the real corpora they are built for, and that gap is worth measuring directly. The corpus (1,623 works, 45,293 fragments, 2,768 knowledge-graph edges), code, and evaluation scripts are released under open licences to support future research on legal NLP for small jurisdictions.

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